1.16 Graph Sketching: Quartic Graphs

Worked Solutions

1a)

$$y = (x-1)^4$$
$$\Rightarrow y = x^4 - 4x^3 + 6x^2 - 4x + 1$$

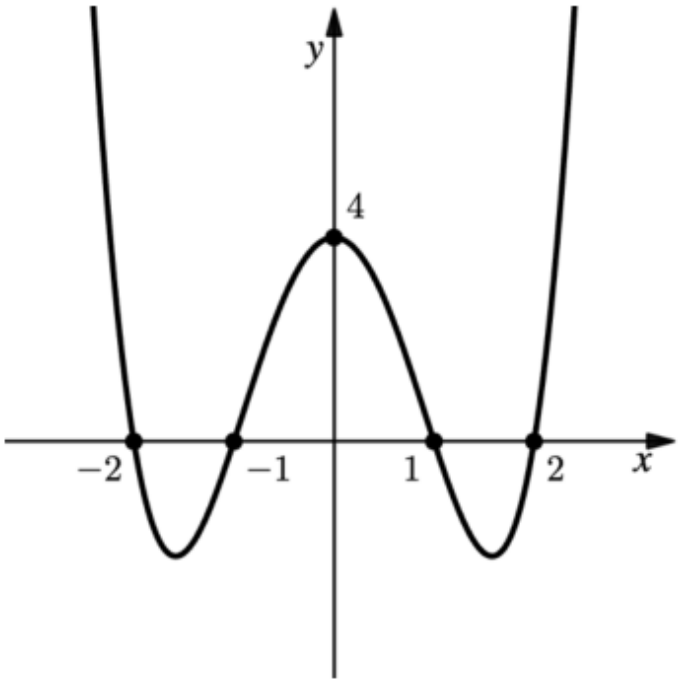
1b)

$$y = (x+2)^4 + 2$$
$$\Rightarrow y = x^4 + 8x^3 + 24x^2 + 32x + 18$$

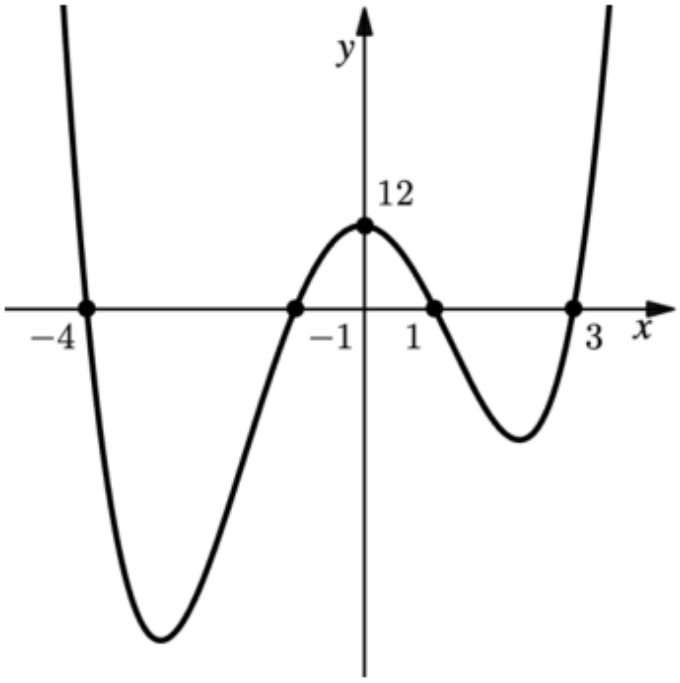
1c)

$$y = (x-3)^4 - 81$$
$$\Rightarrow y = x^4 - 12x^3 + 54x^2 - 108x$$

2a)



2b)



3a)

$$y = 1 - (x+1)^4$$
$$\Rightarrow y = -x^4 - 4x^3 - 6x^2 - 4x$$

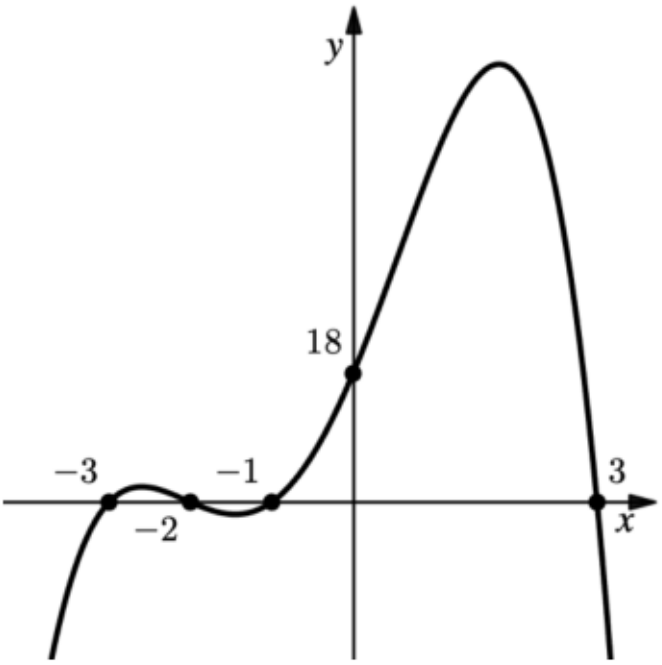
3b)

$$y = 2 - (x-1)^4$$
$$\Rightarrow y = -x^4 + 4x^3 - 6x^2 + 4x + 1$$

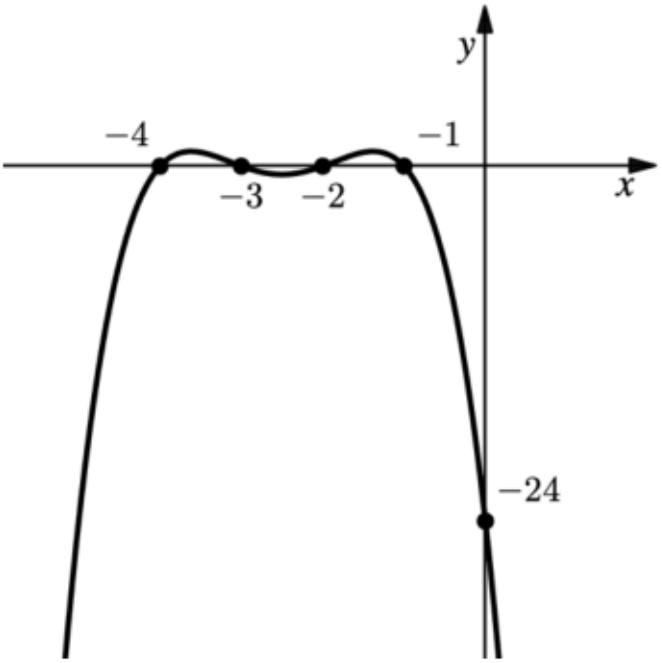
3c)

$$y = -(x+3)^4 - 19$$
$$\Rightarrow y = -x^4 - 12x^3 - 54x^2 - 108x - 100$$

4a)



4b)



5a)

$$y = (x+1)(x-1)(x-2)(x-3)$$

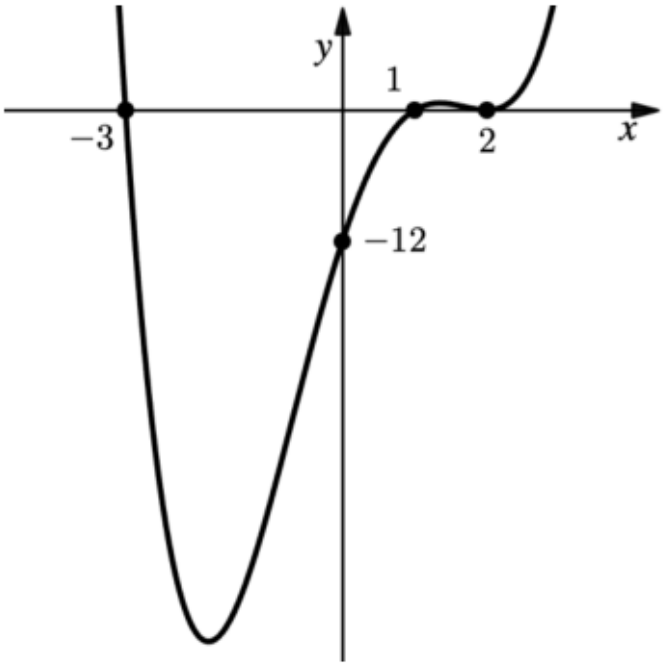
5b)

$$y = (x+3)(x+2)(x-1)(x-2)$$

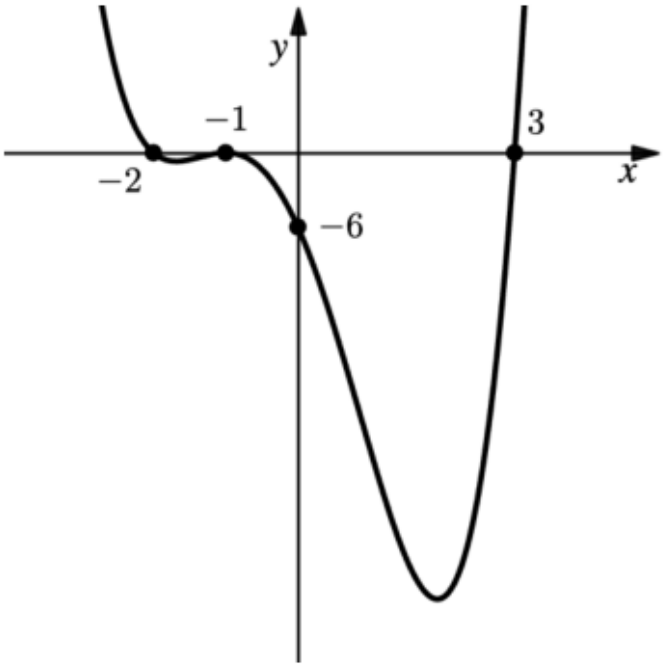
5c)

$$y = (x+1)(x-2)(x-3)(x-4)$$

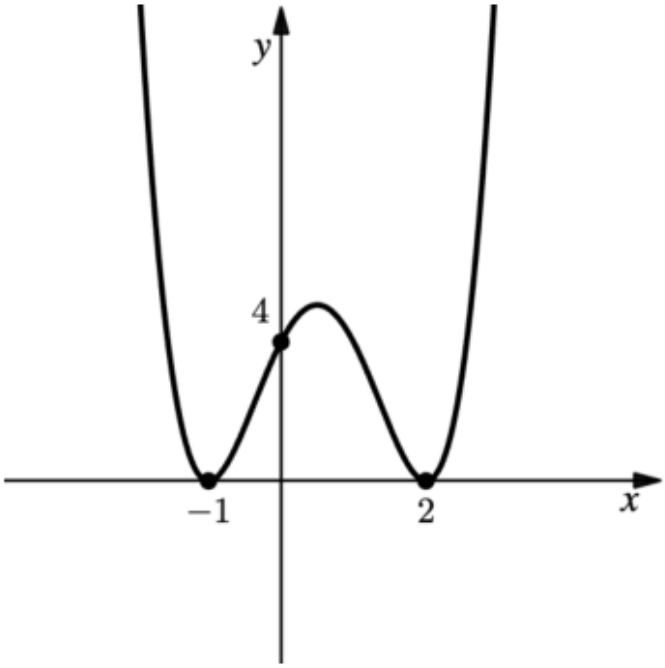
6a)



6b)



6c)



7a)

$$y = (x-1)(x-2)(x+1)^2$$

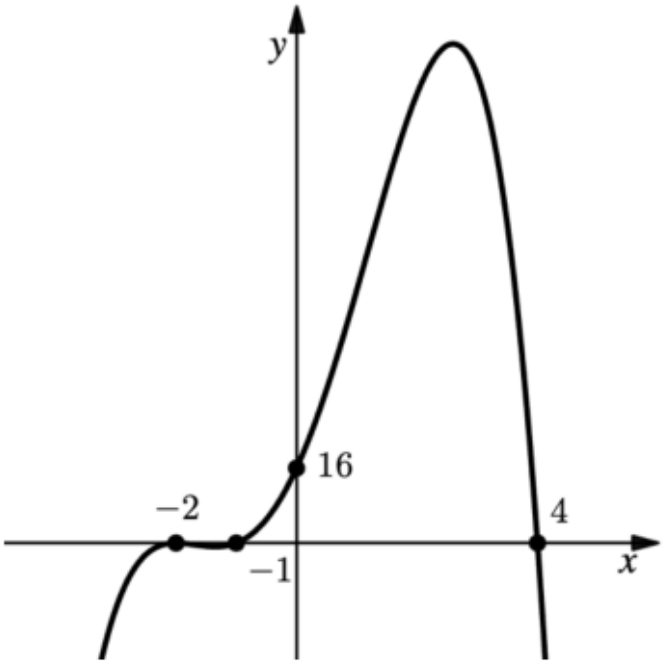
7b)

$$y = (x+3)(x+1)(x-2)^2$$

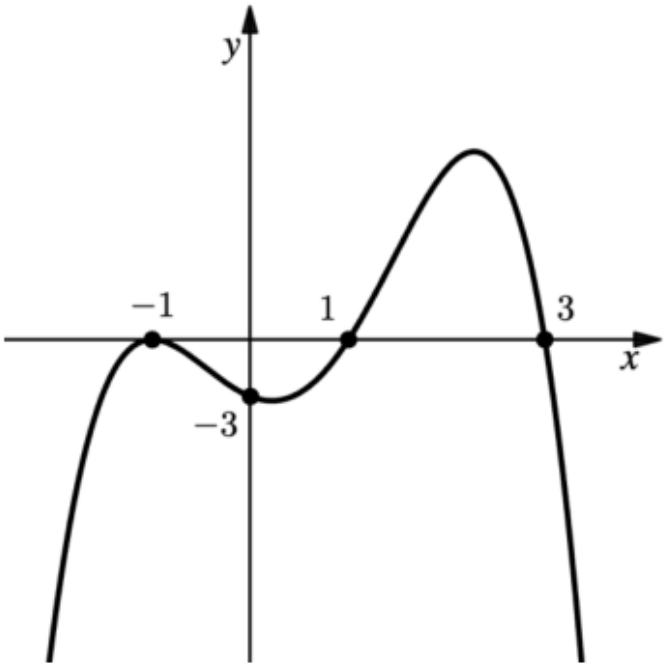
7c)

$$y = (x+3)(x-3)(x+1)^2$$

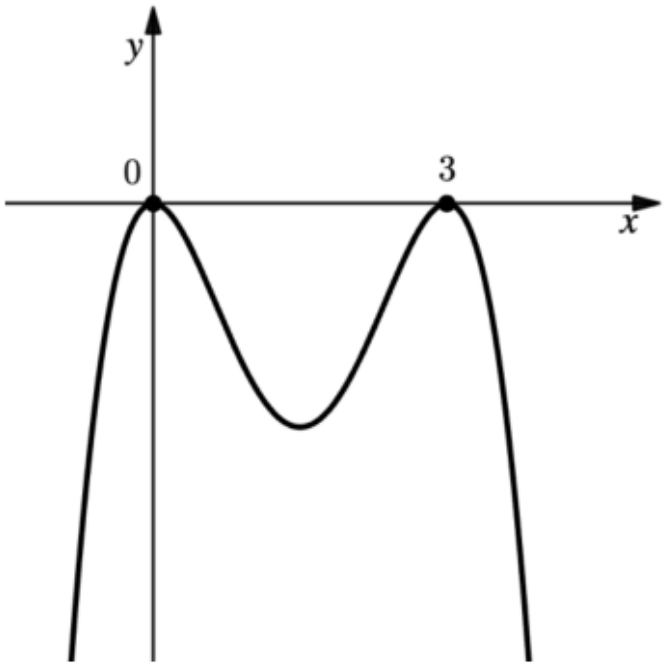
8a)



8b)



8c)



9a)

$$y = 2(x+2)(x+1)(x-2)(x-3)$$

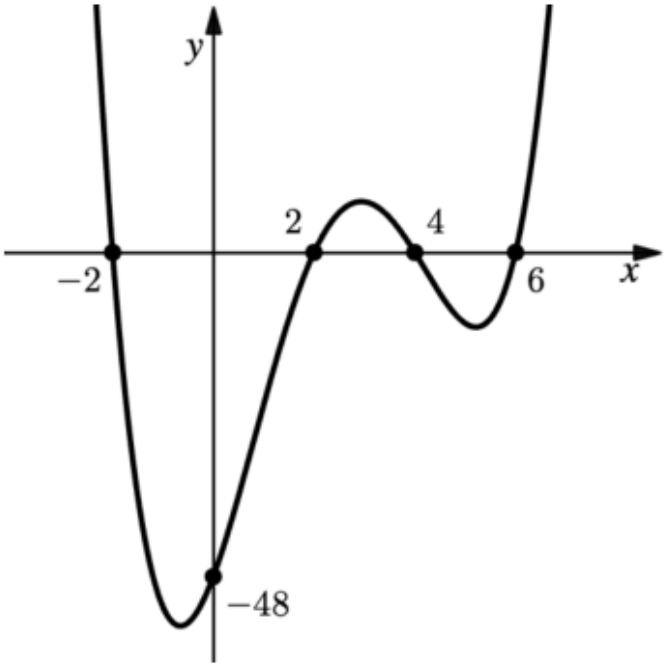
9b)

$$y = -3(x-1)(x-2)(x-3)(x-4)$$

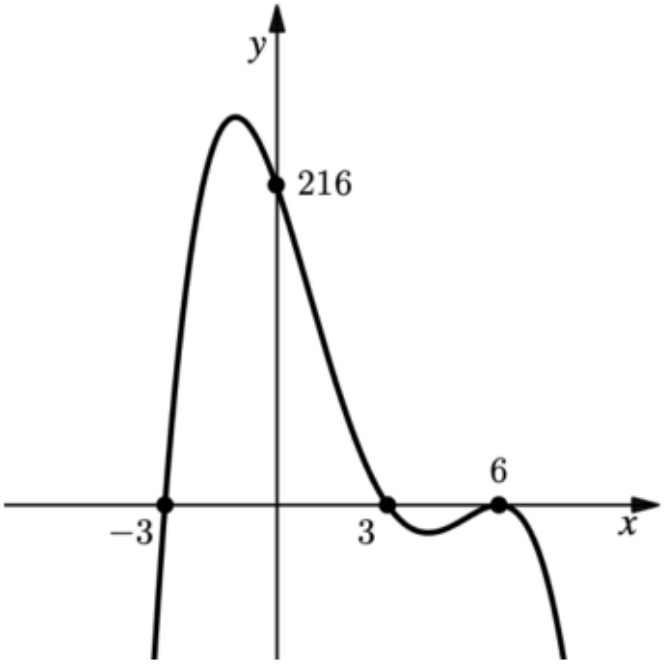
9c)

$$y = -\tfrac{1}{4}(x+4)(x+1)(x-3)(x-5)$$

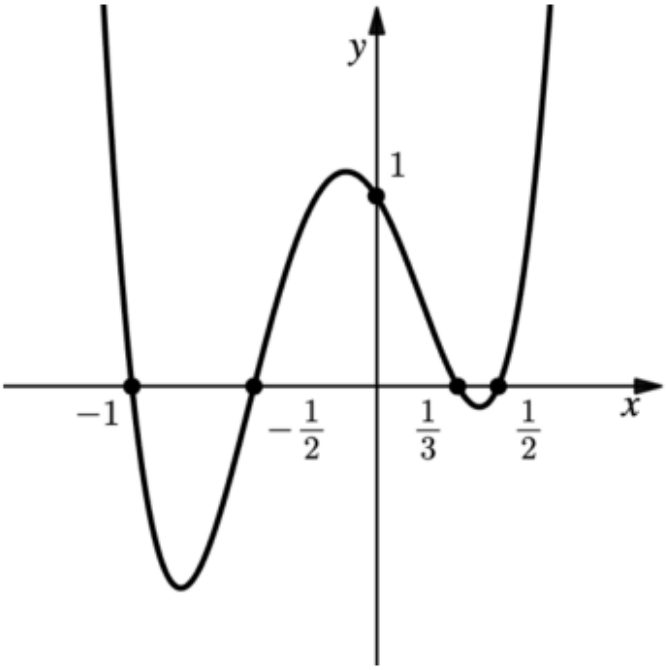
10a)



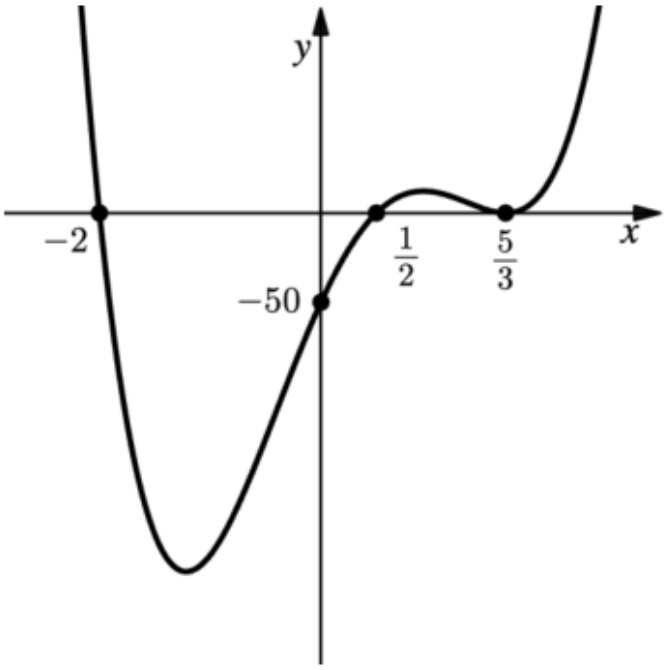
10b)



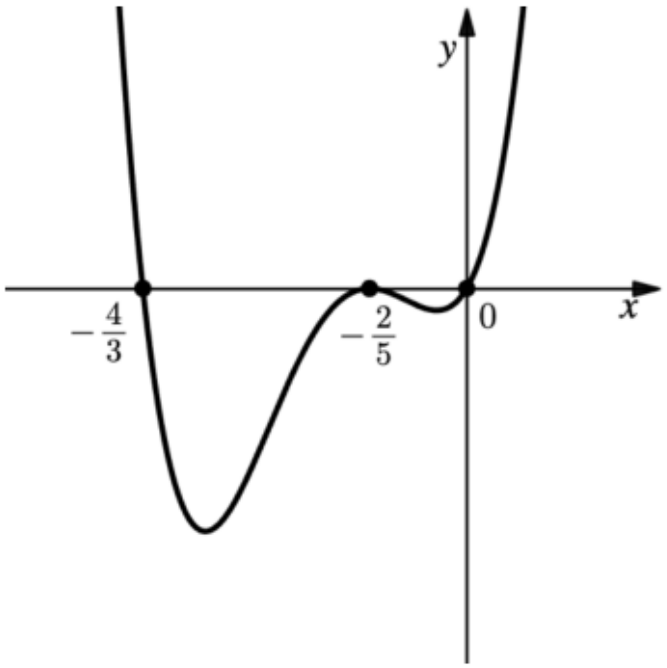
11a)



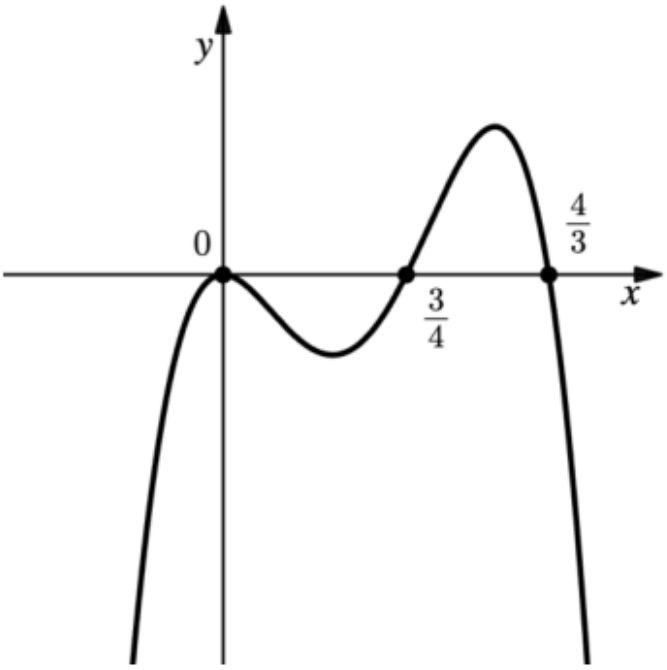
11b)



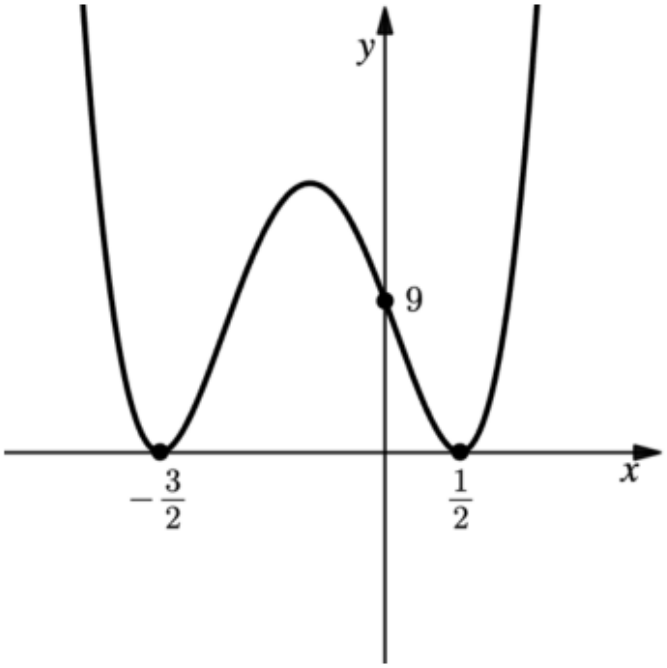
11c)



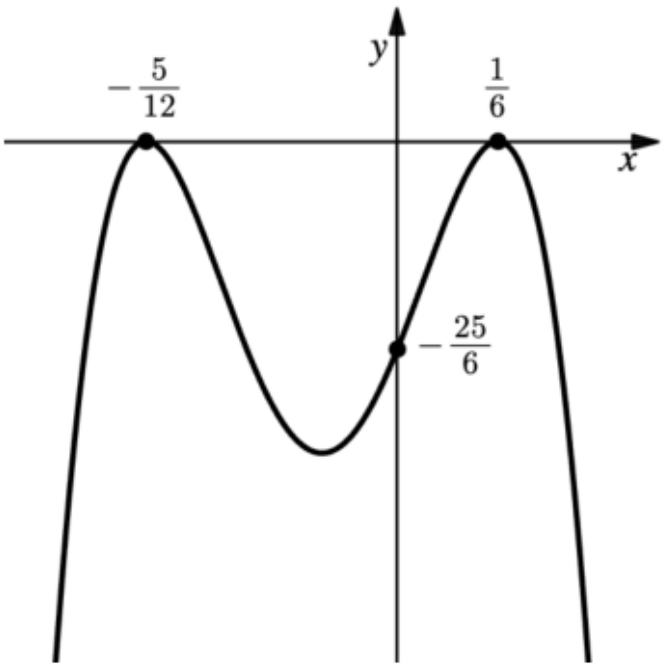
11d)



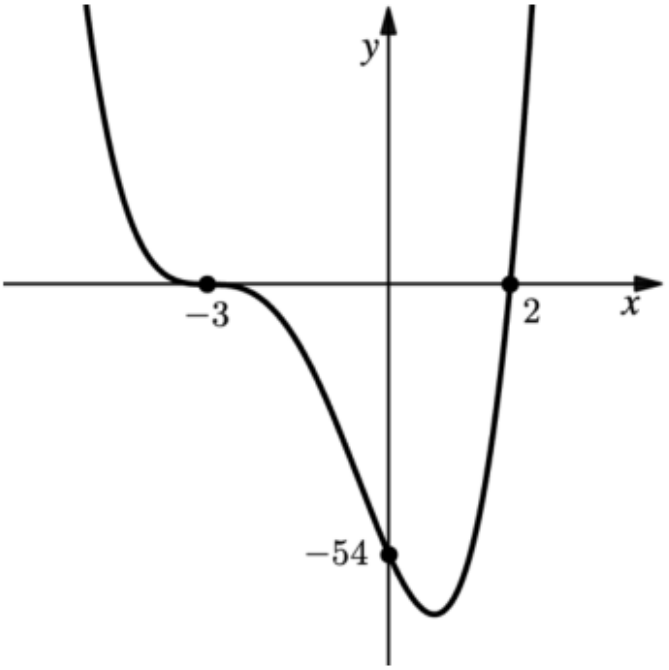
11e)



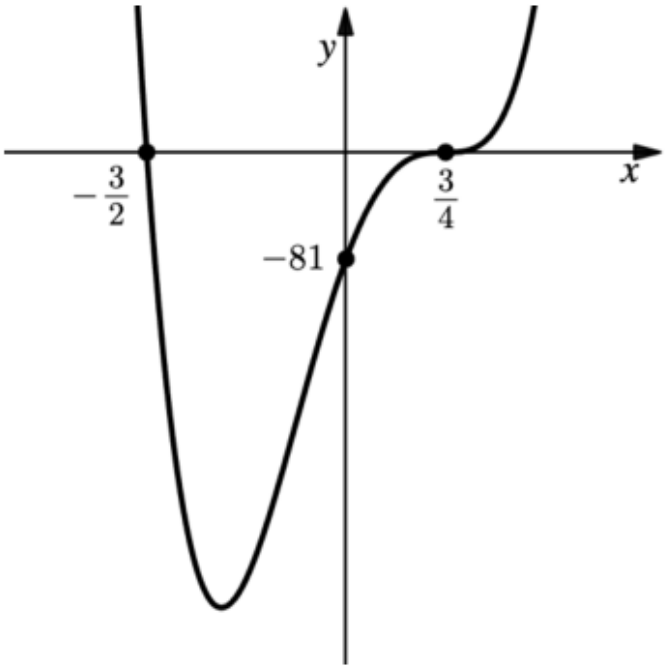
11f)



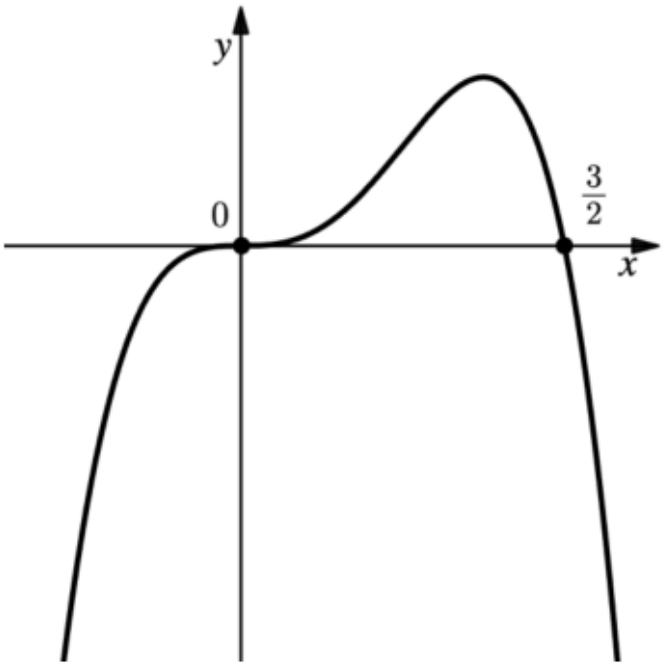
12a)



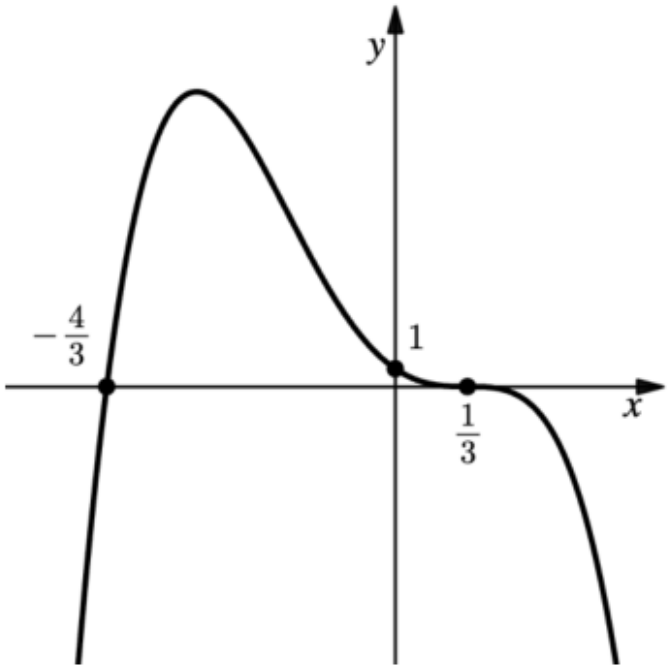
12b)



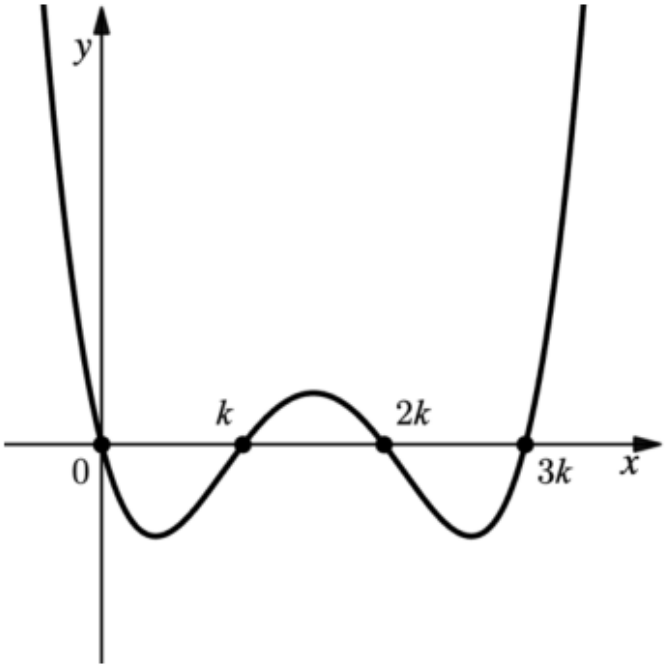
12c)



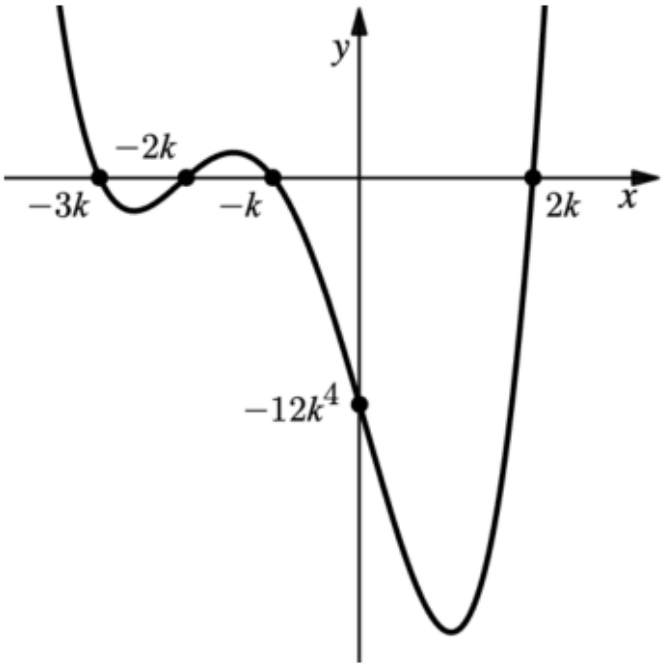
12d)



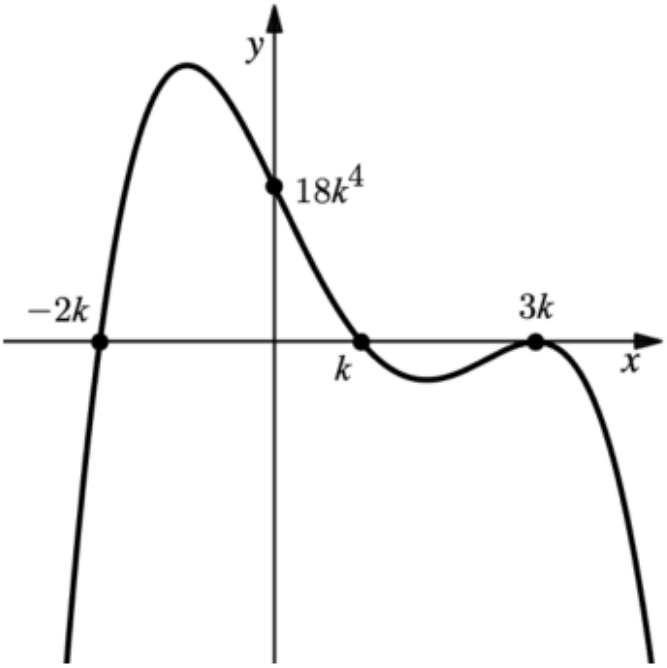
13a)



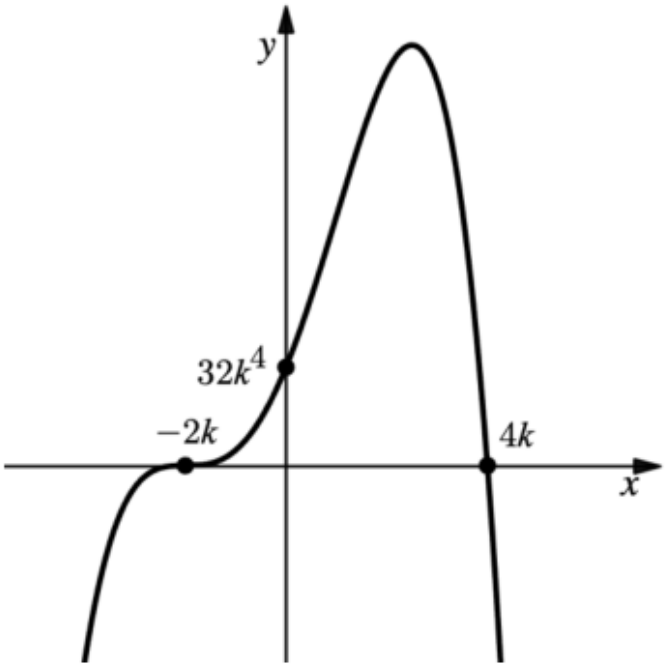
13b)



13c)



13d)



14)

$$0 = \left(x + \frac{k}{2}\right)\left(x - \frac{k}{3}\right)\left(x - \frac{2}{k}\right)(x - 4k)$$

$$\Rightarrow 0 = (2x + k)(3x - k)(kx - 2)(x - 4k)$$

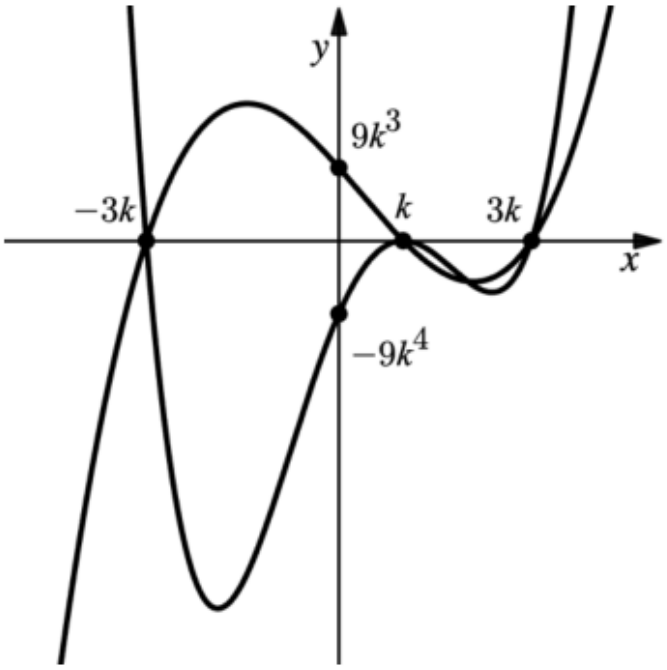
$$y = a(2x + k)(3x - k)(kx - 2)(x - 4k)$$

When $x = 0$, $-8k = a(k)(-k)(-2)(-4k)$

$$\Rightarrow -8k = -8k^3 a \Rightarrow a = \frac{1}{k^2}$$

$$\therefore y = \frac{1}{k^2}(2x + k)(3x - k)(kx - 2)(x - 4k)$$

15)



$$(x-3k)(x+3k)(x-k)^2 = (3k-x)(3k+x)(k-x)$$
$$\Rightarrow (x-3k)(x+3k)(x-k)^2 = (x-3k)(x+3k)(x-k)$$
$$\Rightarrow x-k=1 \Rightarrow x=k+1$$

When $x = k+1$,

$$\begin{aligned} y &= (3k-(k+1))(3k+(k+1))(k-(k+1)) \\ &= (2k-1)(4k+1)(-1) \\ &= -(8k^2-2k-1) \end{aligned}$$

So the curves intersect at $(-3k, 0)$, $(k, 0)$, $(3k, 0)$ and $(k+1, -8k^2+2k+1)$.